Tables of the Thesis

Financial Intermediation and Asset Price Dynamics in the Post-COVID Monetary Expansion. Master Thesis, Chirag Khanna.

This document was written on 2026-10-15 10:13 by `scripts/reproduce_thesis_tables.py`. It contains every table of the thesis, produced from the canonical pipeline outputs in `results/final/`, in the order in which the tables appear in the thesis and with the same numbering. The manifest next to this file records the source files behind each table.

Table 4.1: Descriptive statistics for the shock and the canonical response variables

Variable	*N*	Mean	SD	Min	Max
Shock and response variables					
ECB easing surprise	249	0.046	2.414	−21.122	13.661
house-purchase lending growth	250	3.620	3.173	−0.559	12.174
pure new house-purchase loans	99	10.920	0.189	10.518	11.310
mortgage lending spread	250	2.249	0.906	−0.240	3.752
NFC lending spread	250	2.104	0.451	1.020	3.125
NFC credit	250	15.328	0.110	14.977	15.480
real DAX	250	8.997	0.329	8.185	9.653
ECB assets	250	14.941	0.667	13.704	15.994
real wage tracker excl. one-offs	154	0.118	2.041	−6.926	2.764
German industry real wage bill growth	238	0.408	3.295	−13.373	7.112
employment expectations	250	100.135	10.012	50.800	116.200
Control variables					
EA20 annual HICP inflation	238	2.099	1.968	−0.619	10.102
Deposit facility rate, monthly average	250	0.773	1.415	−0.500	4.000

Notes: Monthly observations, January 2005 through October 2025. Units follow the variable definitions in this chapter. Sample sizes differ because the pure-new-loan, wage-tracker, and wage-bill series start later than the panel. The local projections also control for two lags of the response variable itself. Source: own calculations.

Table 4.2: Pairwise correlations among the shock and the canonical response variables

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) ECB easing surprise										
(2) house-purchase lending growth	0.06									
(3) pure new house-purchase loans	0.07	0.69*								
(4) mortgage lending spread	0.08	0.06	0.53*							
(5) NFC lending spread	0.10	0.12*	0.51*	0.91*						
(6) NFC credit	−0.11	−0.67*	−0.06	−0.36*	−0.39*					
(7) real DAX	−0.06	−0.29*	0.10	−0.72*	−0.76*	0.46*				
(8) ECB assets	−0.14*	−0.43*	0.23*	−0.61*	−0.70*	0.71*	0.82*			
(9) real wage tracker excl. one-offs	0.12	−0.60*	−0.18	−0.02	0.00	−0.29*	0.03	−0.50*		
(10) German industry real wage bill growth	0.13*	−0.03	−0.15	0.14*	0.25*	−0.33*	−0.02	−0.26*	0.36*	
(11) employment expectations	0.02	0.35*	0.04	−0.36*	−0.32*	−0.16*	0.47*	0.23*	−0.43*	0.39*

Notes: Pearson correlations on pairwise complete monthly observations. * $p < 0.05$. Sample sizes vary across pairs because several series start later than the panel. Source: own calculations.

Table 5.1: Ordered timing associations along the intermediation chain

Upstream	Downstream	h_u	h_d	λ	p
mortgage lending spread	house-purchase lending growth	1	6	−1.470	0.088
NFC lending spread	NFC credit	1	6	0.028	0.077
ECB assets	NFC credit	1	3	0.054	0.002
household credit	house-purchase lending growth	1	3	98.465	0.000
house-purchase lending growth	pure new house-purchase loans	1	3	0.152	0.000
NFC credit	real wage tracker excl. one-offs	2	6	2.331	0.855
employment expectations	real wage tracker excl. one-offs	1	6	−0.037	0.042
real DAX	real wage tracker excl. one-offs	1	6	−2.588	0.073

Notes: λ is the association between an earlier upstream movement and a later downstream movement, holding the policy-news shock fixed. This is ordered timing evidence and not structural mediation. Source: own calculations.

Table 5.2: Main synthesis table for persistence evidence

Variable	Sign	Peak horizon	Cumulative persistence	Stability	Interpretation
House-purchase lending growth	Positive	h6	Positive h24 after a partial reversal from h12, 90% HAC CI excludes zero, bootstrap through h12	High: sign 0.86, cumulative 1.00	Core housing-credit response and main empirical contribution.
Mortgage spreads	Positive	h12	Positive h24, 90% CI excludes zero (HAC and moving block)	Very high: sign 1.00, cumulative 1.00	Stable lending-condition margin supporting intermediation.
NFC spreads	Positive	h12	Positive h24, 90% HAC CI excludes zero, bootstrap at h12 only	High: sign 0.90, cumulative 1.00	Smaller but durable corporate credit-price adjustment.
NFC credit	Mixed	h24	Negative h24, CI crosses zero	Weak: sign 0.52, cumulative 0.50	Productive-credit quantity response too imprecise for a persistence claim.
DAX	Negative	h24	Negative h24, 90% CI excludes zero (HAC and bootstrap)	High: sign 0.90, cumulative 1.00	Information-sensitive market response, not asset-appreciation evidence.
ECB assets	Negative	h24	Negative h24, CI crosses zero	Moderate: sign 0.79, cumulative 0.86	Liquidity-stock context, not a strong persistence pillar.
Wage tracker	Mixed	h24	Negative h24, CI crosses zero	Moderate: sign 0.67, cumulative 0.86	Euro-area compensation response attenuates by h24.
German wage bill	Positive	h1	Positive h24, 90% CI excludes zero (HAC and both bootstraps)	High: sign 0.81, cumulative 1.00	Sectoral compensation response, durable through h24.
Employment expectations	Positive then reverse	h1	Positive h24, CI crosses zero	Moderate: sign 0.76, cumulative 0.79	Short-horizon labor expectations, not durable income transmission.

Notes: Peak horizons, accumulated responses, and stability shares summarize the estimates reported in Tables 5.3 and 5.4. CI refers to the HAC interval under the independence approximation, bootstrap to the joint moving-block and wild intervals in Table 6.4. Source: own calculations.

Table 5.3: Baseline monthly impulse responses to a one-standard-deviation ECB easing surprise

	Horizon in months					
Response	0	1	3	6	12	24
house-purchase lending growth	0.048**	0.054**	0.134***	0.149***	0.099	−0.126
	(0.022)	(0.026)	(0.035)	(0.044)	(0.069)	(0.099)
pure new house-purchase loans	−0.007	0.013	0.001	0.020	−0.001	−0.031
	(0.012)	(0.011)	(0.013)	(0.013)	(0.013)	(0.027)
mortgage lending spread	0.006	0.021**	0.015	0.027	0.050*	0.022
	(0.007)	(0.008)	(0.016)	(0.020)	(0.026)	(0.023)
NFC lending spread	0.000	0.017*	0.003	0.015	0.025**	0.010
	(0.007)	(0.009)	(0.009)	(0.011)	(0.012)	(0.011)
NFC credit	0.000	−0.001	0.000	0.000	0.000	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
real DAX	−0.001	−0.002	−0.006	−0.009	−0.021**	−0.027**
	(0.005)	(0.006)	(0.005)	(0.008)	(0.010)	(0.013)
ECB assets	−0.004	−0.002	−0.003	−0.003	−0.010	−0.011
	(0.003)	(0.004)	(0.006)	(0.007)	(0.012)	(0.015)
real wage tracker excl. one-offs	−0.069*	−0.020	−0.081**	−0.089	−0.042	0.260
	(0.037)	(0.042)	(0.041)	(0.062)	(0.061)	(0.218)
German industry real wage bill growth	0.205**	0.422	0.347*	0.403***	0.216	−0.105
	(0.096)	(0.270)	(0.191)	(0.118)	(0.175)	(0.117)
employment expectations	0.408	1.173	0.627	0.336	0.217	−0.800**
	(0.287)	(1.020)	(0.471)	(0.327)	(0.437)	(0.369)

Notes: Local-projection coefficients scaled to a one-standard-deviation easing surprise. HAC (Newey–West) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own calculations.

Table 5.4: Accumulated responses at the medium and long horizons

	Horizon in months		
Response	6	12	24
house-purchase lending growth	0.385†	0.483†	0.357†
pure new house-purchase loans	0.027	0.025	−0.006
mortgage lending spread	0.068†	0.118†	0.140†
NFC lending spread	0.035†	0.060†	0.071†
NFC credit	−0.001	−0.001	−0.002
real DAX	−0.019	−0.040†	−0.067†
ECB assets	−0.012	−0.022	−0.032
real wage tracker excl. one-offs	−0.260†	−0.302†	−0.041
German industry real wage bill growth	1.377†	1.593†	1.488†
employment expectations	2.545†	2.762†	1.962

Notes: Running sum of the normalized impulse responses through each horizon. Intervals combine the horizon-level HAC standard errors under an independence approximation across horizons. † marks accumulated responses whose 90 percent HAC interval excludes zero. Table 6.4 reports the dependence-aware bootstrap intervals for the same accumulated responses. Source: own calculations.

Table 5.5: Empirical assessment of the three thesis hypotheses

Hypothesis	Evidence	Assessment
H1: Housing-credit persistence	House-purchase lending growth is positive on impact and peaks at h6. Its accumulated response excludes zero from h0 through h12 under the HAC interval and both bootstraps.	Supported at short and medium horizons.
H2: Compensation-linked attenuation	The canonical wage tracker moves at short and medium horizons but loses h24 cumulative exclusion, while employment expectations reverse. The German industry wage bill, a sectoral robustness proxy, keeps its sign and excludes zero from h1 through h24 under every interval method, which is the proxy boundary on this verdict.	Supported with proxy boundary.
H3: Differential financial-intermediation transmission	Housing-credit growth and lending spreads are stronger and more persistent than NFC credit quantities and the euro-area wage tracker.	Conditionally supported.

Notes: H1 to H3 are Hypotheses 1 to 3 of Chapter 1, derived in Section 3.4. CI refers to the HAC interval under the independence approximation, bootstrap to the joint moving-block and wild intervals in Table 6.4. Source: own assessment based on own calculations.

Table 6.1: Information-effect classification of ECB announcement events

Event classification	Events	Share (%)
possible pure monetary shock	86	35.2
mixed sign factor yield conflict	81	33.2
possible information shock	68	27.9
ambiguous zero response	9	3.7
Total	244	100.0

Notes: Events are classified by comparing the sign of the rate factor with the yield and equity responses in the announcement window. Counts cover the full event archive. Source: own calculations.

Table 6.2: Monthly first-stage relevance of the strongest candidate policy bridges

Surprise factor	Policy target	F	Partial R^2	N	Screen
target factor	deposit facility rate, end of month	17.07	0.066	246	fragile candidate
timing factor	ECB assets, log monthly average	10.99	0.044	244	fragile candidate
fg factor	deposit facility rate, monthly average	9.43	0.038	244	fragile candidate
timing factor	ECB assets, log end-of-month stock	7.05	0.029	244	fragile candidate
qe factor	deposit facility rate, end of month	5.73	0.040	142	fragile candidate
target factor	ECB assets, log monthly average	4.42	0.018	246	reject weak relevance
target factor	household credit	3.02	0.012	246	reject weak relevance
qe factor	NFC credit	2.50	0.018	142	reject weak relevance

Notes: Eight strongest bridges from the monthly first-stage tournament, ranked by F-statistic. No candidate clears the strength and stability screen jointly, which is why the thesis reports reduced-form responses to policy news. Source: own calculations.

Table 6.3: Residual diagnostics for the baseline local projections

		Horizons (months) at which the test rejects at 5 percent			
Response	N	Serial corr.	Heterosk.	ARCH	Non-normal
house-purchase lending growth	246	all	1, 3, 6, 12, 24	all	0, 1, 3
pure new house-purchase loans	97	all	3, 6	12	none
mortgage lending spread	246	6, 12, 24	1, 3, 6, 12, 24	all	all
NFC lending spread	246	0, 12, 24	1	all	all
NFC credit	246	all	6, 12, 24	1, 3, 6, 12, 24	0, 1, 3, 6, 12
real DAX	246	6, 12, 24	all	1, 3, 6, 12, 24	all
ECB assets	246	all	0, 1, 3	all	0, 1, 3, 24
real wage tracker excl. one-offs	152	0, 1, 3, 6, 12	0, 1, 6, 12, 24	1, 3, 6, 12, 24	all
German industry real wage bill growth	235	0, 1, 3, 24	1, 3	all	all
employment expectations	246	3, 6, 12	0, 1	1, 3, 6, 12, 24	all

Notes: Baseline target-shock regressions, full sample. N is the estimation sample at $h = 0$. Serial corr.: Ljung–Box test on the twelve residual autocorrelations at lags $h + 2$ to $h + 13$, just beyond the Newey–West bandwidth $h + 1$. Heterosk.: Koenker's studentized Breusch–Pagan test of the squared residuals on the regressors. ARCH: Engle's LM test with twelve lags. Non-normal: Jarque–Bera. "all" means every one of the six horizons. Source: own calculations.

Table 6.4: Accumulated responses: 90 percent intervals from the dependence-aware bootstraps

		Horizon in months		
Response	Bootstrap	6	12	24
house-purchase lending growth	moving block	[0.167, 0.545]†	[0.109, 0.731]†	[-0.078, 0.734]
	wild	[0.143, 0.644]†	[0.151, 0.819]†	[-0.063, 0.809]
pure new house-purchase loans	moving block	[-0.042, 0.082]	[-0.065, 0.091]	[-0.088, 0.073]
	wild	[-0.059, 0.101]	[-0.068, 0.107]	[-0.085, 0.063]
mortgage lending spread	moving block	[0.016, 0.134]†	[0.039, 0.218]†	[0.034, 0.267]†
	wild	[-0.009, 0.148]	[-0.001, 0.234]	[-0.016, 0.293]
NFC lending spread	moving block	[-0.004, 0.081]	[0.008, 0.125]†	[-0.011, 0.155]
	wild	[-0.031, 0.106]	[-0.024, 0.149]	[-0.040, 0.176]
NFC credit	moving block	[-0.005, 0.003]	[-0.008, 0.006]	[-0.013, 0.009]
	wild	[-0.008, 0.006]	[-0.012, 0.010]	[-0.018, 0.013]
real DAX	moving block	[-0.044, 0.011]	[-0.075, 0.000]	[-0.111, -0.022]†
	wild	[-0.044, 0.010]	[-0.074, -0.006]†	[-0.123, -0.011]†
ECB assets	moving block	[-0.035, 0.015]	[-0.058, 0.021]	[-0.084, 0.026]
	wild	[-0.049, 0.028]	[-0.078, 0.040]	[-0.114, 0.058]
real wage tracker excl. one-offs	moving block	[-0.530, 0.061]	[-0.726, 0.175]	[-0.578, 0.533]
	wild	[-0.475, -0.014]†	[-0.572, -0.001]†	[-0.362, 0.332]
German industry real wage bill growth	moving block	[0.626, 2.136]†	[0.721, 2.489]†	[0.536, 2.458]†
	wild	[0.403, 2.356]†	[0.404, 2.730]†	[0.246, 2.691]†
employment expectations	moving block	[1.018, 4.632]†	[0.579, 5.403]†	[-0.462, 4.863]
	wild	[-0.365, 6.132]	[-0.810, 7.365]	[-1.582, 6.679]

Notes: Percentile intervals for the running sum of the normalized responses to the target easing surprise from 399 joint replications that resample the residuals of all six horizons together, in moving blocks of twelve months or with wild sign flips. † marks intervals that exclude zero. Compare with the HAC intervals under the independence approximation in Table 5.4. Source: own calculations.

Table A.1: Peak response magnitude and timing

Response	Peak h	Peak	90% interval		p	N
house-purchase lending growth	6	0.149	[0.077,	0.221]	0.001	240
pure new house-purchase loans	24	−0.031	[−0.075,	0.014]	0.253	73
mortgage lending spread	12	0.050	[0.007,	0.093]	0.054	234
NFC lending spread	12	0.025	[0.006,	0.045]	0.034	234
NFC credit	24	−0.001	[−0.004,	0.002]	0.561	222
real DAX	24	−0.027	[−0.049,	−0.006]	0.037	222
ECB assets	24	−0.011	[−0.035,	0.014]	0.476	222
real wage tracker excl. one-offs	24	0.260	[−0.099,	0.620]	0.233	128
German industry real wage bill growth	1	0.422	[−0.023,	0.867]	0.119	234
employment expectations	1	1.173	[−0.505,	2.851]	0.250	245

Notes: Largest absolute normalized response across the reported horizons, with its HAC 90 percent interval and estimation sample size. Source: own calculations.

Table A.2: Data sources, series codes, and access dates

Variable	Source and series code	Accessed
Policy shock and announcement surprises		
Target, timing, FG, and QE factors	Ragusa (n.d.), factor vintage of 30 Oct 2025, built with the method of Altavilla et al. (2019)[a]	15 May 2026
Announcement-window OIS and STOXX 50 surprises	European Central Bank (n.d.-d)[a]	15 May 2026
Policy rate and liquidity		
Deposit facility rate, daily	ECB, `FM.D.U2.EUR.4F.KR.DFR.LEV`, retrieved as `ECBDFR` (European Central Bank, n.d.-c)[a]	15 May 2026
Eurosystem total assets, weekly	ECB, `ILM.W.U2.C.T000000.Z5.Z01`, retrieved as `ECBASSETSW` (European Central Bank, n.d.-a)[a]	15 May 2026
Wu–Xia shadow rate	Wu and Xia (n.d.), method of Wu and Xia (2020)[a]	15 May 2026
Lending conditions and credit (European Central Bank, n.d.-b)		
Mortgage lending rate	`MIR.M.U2.B.A2C.A.R.A.2250.EUR.N`[a]	16 May 2026
NFC lending rate	`MIR.M.U2.B.A2A.A.R.A.2240.EUR.N`[b]	16 May 2026
House-purchase lending growth	`BSI.M.U2.Y.U.A22.A.I.U2.2250.Z01.A`[a]	16 May 2026
Pure new house-purchase loans	`MIR.M.U2.B.A2C.A.B.A.2250.EUR.P`[a]	16 May 2026
Household loan stock	`BSI.M.U2.N.A.A20T.A.1.U2.2250.Z01.E`[b]	15 May 2026
NFC loan stock	`BSI.M.U2.N.A.A20T.A.1.U2.2240.Z01.E`[b]	15 May 2026
Bank Lending Survey standards and demand	`BLS.Q.U2.ALL.{Z.H.H, O.E.Z, Z.H.C}.B3.{ST.S, ZZ.D}.WFNET`[a]	16 May 2026
Asset prices and deflators		
DAX close, monthly	Yahoo Finance (n.d.), `^GDAXI`[a]	15 May 2026
German residential property prices	BIS, `Q:DE:N:628`, retrieved as `QDEN628BIS` (Bank for International Settlements, n.d.)[b]	15 May 2026
HICP, Germany, EA20, and euro area	Eurostat (n.d.-b), index 2005 = 100, all items[a]	15 May 2026
Compensation and labour market		
ECB wage tracker (ex one-offs, headline, unsmoothed)	`EWT.M.U2.N.WT.{INWX, INWS, INWR}._T.4F0.GY` (European Central Bank, n.d.-b)[b]	16 May 2026
Compensation per employee, EA20	`MNA.Q.Y.I9.W2.S1.S1._Z.COM_PS._Z._T._Z.IX.V.N` (European Central Bank, n.d.-b)[b]	15 May 2026
Consumer income and unemployment expectations	`CES.M.Z18.ALL.T.{C3220, C4031}.NUM_VAR.WM` (European Central Bank, n.d.-b)[a]	16 May 2026
German industry wage bill	`sts_inlb_m`, DE, B–E36, NSA (Eurostat, n.d.-a)[a]	16 May 2026
German retail trade volume	`sts_trtu_m`, DE, volume of sales (Eurostat, n.d.-a)	15 May 2026
Employment expectations, EA20	`ei_bsee_m_r2`, EEI and services (Eurostat, n.d.-a)[a]	16 May 2026
Unemployment rate, EA21	`une_rt_m`, seasonally adjusted (Eurostat, n.d.-a)[b]	16 May 2026

Notes: [a] Identical to the live source, at the precision stored in the raw file, when re-checked on 27 September 2026. [b] Same series, revised by the provider after the access date. The thesis uses the vintage downloaded on the access date. Source: own compilation.

Table A.3: Canonical pipeline artifacts underlying the reported results

Artifact	Path
Canonical monthly model dataset	data/processed/eu_de/final_monthly_model_dataset.csv
Monthly variable dictionary	data/processed/eu_de/monthly_variable_dictionary.csv
Information-effect event screen	results/final/diagnostics/information_effect_event_screen.csv
Instrument strength matrix	results/final/diagnostics/instrument_strength_matrix.csv
Monthly first-stage tournament	results/final/diagnostics/monthly_first_stage_tournament.csv
Banking timing matrix	results/final/mechanism/banking_timing_matrix.csv
Sequential transmission outputs	results/final/mechanism/sequential_timing_outputs.csv
Regime-specific reduced-form LP output	results/final/regime/monthly_regime_reduced_form_lp.csv
OLS policy-rate direction check	results/final/robustness/monthly_ols_policy_rate_comparison.csv
Cumulative stability outputs	results/final/stability/monthly_cumulative_stability_long.csv
Directional stability matrix	results/final/stability/monthly_directional_stability_matrix.csv
Rolling and recursive LP stability	results/final/stability/monthly_stability_metrics.csv
Cumulative persistence table	results/final/tables/cumulative_persistence_table.csv
Horizon significance table	results/final/tables/horizon_significance_table.csv
Normalized baseline IRF output	results/final/tables/normalized_irf_outputs.csv
Peak response table	results/final/tables/peak_response_table.csv
Stability metrics	results/final/tables/stability_metrics.csv
Bootstrap sensitivity summary	results/final/uncertainty/bootstrap_sensitivity_summary.csv
Horizon dependence diagnostics	results/final/uncertainty/horizon_dependence_diagnostics.csv

Source: own compilation.

Table A.4: Software, tools and data portals used in the preparation of this thesis

Tool	Version	Use
Python	3.13.11	Shock construction, local projections, bootstraps, diagnostics, thesis tables and figures
numpy, pandas, scipy	2.4.1, 3.0.0, 1.17.0	Numerical work and data handling
statsmodels	0.14.6	Regression estimation with HAC covariance
matplotlib	3.10.8	Figures
R	4.4.2	Construction of the processed datasets, two supplementary figures
readxl, dplyr, tidyr	1.4.3, 1.1.4, 1.3.1	Data import and transformation
jsonlite, ggplot2	1.8.9, 3.5.1	Configuration files and plotting
XeLaTeX (TeX Live 2026), biblatex with biber	biber 2.21	Typesetting and reference management
Data portals		
ECB Data Portal (data.ecb.europa.eu)		
Eurostat Database (ec.europa.eu/eurostat)		
FRED, Federal Reserve Bank of St. Louis (fred.stlouisfed.org)		
ECB Euro Area Monetary Policy Event-Study Database (ecb.europa.eu)		
Monetary policy factors (gragusa.org/factors)		
Wu-Xia shadow rate (sites.google.com/view/jingcynthiawu)		
Yahoo Finance (finance.yahoo.com)		

Source: own compilation.